P8139 Final Project Proposal: methylation-driven transcriptional silencing in TCGA-KIRC

Yongyan Liu (yl6107) — Homework #5

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1 Cancer type and research question

Cancer type. Clear-cell renal cell carcinoma (ccRCC), TCGA project KIRC. ccRCC is chosen because it is the largest paired-design cohort in TCGA, its molecular biology is dominated by VHL inactivation (a well-characterized positive control), and its single dominant histology limits biological heterogeneity (TCGA Research Network 2013; Kaelin 2007).

Motivating problem. Paired tumor–adjacent-normal methylation analyses on the Illumina 450K platform routinely declare more than half of tested CpGs differentially methylated at FDR < 0.05. The list is statistically valid but biologically uninformative because most differentially methylated CpGs are passenger or cell-composition changes, not regulators of transcription.

Research question. *Which CpG methylation changes drive transcriptional silencing of their cognate gene?* This is the epigenetic counterpart of Knudson's tumor-suppressor second hit and the operationally relevant question for cancer epigenetics. We address it by integrating paired DNA methylation with paired RNA-seq from the same patients, requiring concordant evidence from two statistical tests before declaring a CpG transcriptionally silencing.

2 Data sources, samples, and study design

Data acquisition. DNA methylation 450K β-values, RNA-seq STAR-Counts (used for $\log_2(\text{TPM}+1)$), and harmonized clinical metadata are retrieved from the GDC with `TCGAbiolinks::GDCquery` and `GDCdownload` (Bioconductor 3.22; Colaprico et al. 2016).

Samples and design. This is a **paired case–control design** at the methylation level (each patient contributes a primary-tumor `-01A` aliquot and a solid adjacent-normal `-11A` aliquot) combined with an **independent case-mostly design** at the integration level (most patients have only the tumor RNA-seq). One `-05A` "additional new primary" is excluded by pre-registration. Sample counts:

Table 1: TCGA-KIRC cohort sizes after pre-registered exclusions.

Modality	N
DNA methylation 450K (aliquots)	484 (324 tumor + 160 normal)
DNA methylation 450K (patients)	319
Paired methylation set (both -01A and -11A)	160 patients
RNA-seq STAR-Counts (patient $\times$ sample-type keys)	605
Integration set: matched (patient, sample-type) keys	342 (318 tumor + 24 adjacent-normal)

The DM test runs on the 160 paired patients; the methylation–expression integration test runs on the 342-key matched set.

3 DNA methylation preprocessing steps

The 485,577-probe β-value matrix is filtered sequentially, with attrition recorded. Preprocessing is standard and conservative; no probes are added back at any later stage.

Table 2: Preprocessing attrition for DNA methylation.

Step	Probes retained
Baseline (450K array)	485,577
Intersect with Illumina 450K annotation (drop unmapped)	472,651
Drop chr X / chr Y (avoid sex-driven PC2 in EDA)	461,295
Cross-reactive + SNP-overlapping probes (DMRcate; Chen 2013, Peters 2015)	433,595
Drop probes with > 20% sample missingness	395,325
Complete-case across all 484 samples (paired-test ready)	320,953

β-values are clipped to $[0.001, 0.999]$ and transformed to $M = \log_2(\beta/(1-\beta))$ for variance stabilization in the downstream paired test (Du et al. 2010). RNA-seq counts are converted to TPM and log-transformed as $\log_2(\mathrm{TPM} + 1)$; replicate aliquots within (patient, sample-type) are mean-aggregated.

4 Analysis plan

The plan combines two independent statistical tests; **a CpG is "silencing" only if both tests agree with concordant direction**. This is the mechanistic translation of Knudson's two-hit hypothesis into the methylation–expression coupling regime.

(1) Differential methylation (DM) test — paired moderated t (needed for the methylation-change direction; without DM, MEA alone cannot distinguish promoter silencing from gene-body activation). For each CpG, within-pair M-value differences $D_i = M_i^{\mathrm{tumor}} - M_i^{\mathrm{normal}}$ ($i = 1, \dots, 160$) are tested against $H_0 \colon \mathbb{E}[D] = 0$ using `limma::lmFit` with an intercept-only design and `eBayes` shrinkage (Smyth 2004; Ritchie et al. 2015); for balanced paired data this is statistically equivalent to the moderated paired t-test, and patient-level confounders cancel in D_i.

(2) Methylation–expression association (MEA) test — partial-correlation regression (DM is silent on transcriptional consequence; MEA tests the gene-regulatory effect that turns "differential" into "silencing." Adjusting for `sample_type` is critical because cross-sample correlation is dominated by the bulk tumor/normal mean shift; e.g., VHL cross-cohort $\rho = -0.03$ vs within-status partial $p = 2 \times 10^{-6}$). For each promoter (CpG, gene) pair (annotated as `TSS1500`, `TSS200`, `5'UTR`, or `1stExon`), we fit

$$\log_2(\mathrm{TPM}_{\mathrm{gene}} + 1) \;=\; \alpha + \beta\, \beta_{\mathrm{CpG}} + \gamma^\top Z + \varepsilon, \quad Z = (\text{sample\_type, age, sex, tissue source site}),$$

across the 342 matched samples; the test is computed analytically as a partial correlation after QR residualization on the 15-column design matrix, vectorizing to all 137,451 promoter pairs in <1 second.

(3) Silencing definition (directional concordance of the two independent tests captures the Knudson-second-hit signature and excludes coincidental coupling, gene-body activation, or DM that is statistically real but not gene-regulatory). A (CpG, gene) pair is **silencing** if (i) DM FDR < 0.05, (ii) MEA FDR < 0.05, and (iii) $\widehat{\mathrm{logFC}}_{\mathrm{DM}} > 0$ and $\hat{\beta}_{\mathrm{MEA}} < 0$. The reverse direction (hypomethylation $\cap$ positive coupling) is recorded as **activation** and reported descriptively.

(4) Pattern partition by responder fraction (mean-based DM dilutes signals present in only a minority of tumors; e.g., VHL is silenced via methylation in only ~15% of ccRCC tumors yet is the dominant ccRCC driver — without a per-tumor responder count, this pattern is invisible. The partition is *also* a method validation: if the silencing screen correctly identifies subgroup-driven events, the recovered VHL responder rate should match the documented ~15% literature value, and recovering it is concurrent-validity evidence that the method captures the right kind of heterogeneity). Each silencing CpG is annotated by $\hat{\pi}_{\text{resp}}$, the fraction of tumors whose β deviates from the adjacent-normal mean by > 0.2 (the standard 450K biological-significance threshold). Silencing pairs split into **subgroup-driven** ($\hat{\pi}_{\text{resp}} < 0.3$) and **cohort-wide** ($\geq 0.3$); sensitivity to 0.2 / 0.3 / 0.4 is reported.

(5) Pathway enrichment and cell-composition control — the central methodological claim. *Setup*: bulk tumor DM is confounded by cell composition, because tumor tissue carries more infiltrating leukocytes than adjacent normal kidney; any CpG that distinguishes immune from epithelial cells appears differentially methylated for compositional reasons alone. *Standard fix*: reference-based deconvolution (EpiDISH, CIBERSORT) — adds a tool dependency and requires a tissue-matched reference panel. *Hypothesis*: silencing's directional concordance criterion automatically excises this signal, because an immune CpG bulk-hypomethylated in tumor *by* infiltration also carries elevated bulk mRNA from the same cells — exactly the opposite of silencing's hypermethylation + low-mRNA signature, so it fails directional concordance and never enters the silencing list. *Falsifiable prediction*: immune pathways dominant in raw-DM enrichment should disappear under the silencing restriction. *Implementation*: GSEA on raw-DM rankings (`fgsea`; Korotkevich et al. 2021) is compared against ORA on the silencing gene list (hypergeometric on Hallmark + KEGG Legacy; Liberzon et al. 2015; Subramanian et al. 2005).

(6) Pre-registered validation (without external ground truth, the pipeline could be self-consistent but biologically wrong; pre-registering literature anchors before examining data prevents this). (a) An 18-gene literature TSG list (VHL, RASSF1, SFRP1/2/4/5, DKK3, GATA5, BNC1, COL14A1, PCDH17, SLIT2, CDKN2A, APAF1, TIMP3, NEFH, UCHL1, PITX2) serves as concurrent-validity sanity check. (b) **VHL positive control**: at least one VHL-promoter CpG must flag as silencing. (c) **Calibration**: the DM test's λ_{GC} is paired with a sign-flipping permutation null on 30,000 probes $\times$ 20 replicates. (d) **External replication**: the identical pipeline is run on TCGA-KIRP (papillary RCC, 45 paired patients) and the silencing gene-list overlap with KIRC is tested for hypergeometric enrichment over random sampling. (e) **Threshold sensitivity**: silencing counts and ORA top-hits are reported across a 3×3 (DM FDR, MEA FDR) $\in \{0.01, 0.05, 0.10\}$ grid.

(7) Multiple-testing correction (DM: 320,953 CpGs; MEA: 137,451 pairs; Bonferroni is too conservative given pervasive real signal, but uncorrected p-values yield thousands of false positives). BH FDR at $q < 0.05$ is applied separately to each test; the silencing AND of two independently FDR-controlled sets is conservative relative to either single-test FDR. BH FDR is also used across gene-set families for pathway enrichment. Calibration is established by a sign-flipping permutation null (step (6) above) rather than by Bonferroni.

5 Figures, tables, and preliminary findings (pipeline complete)

Planned outputs. Six figures: (1) two-panel PCA on top-variance CpGs showing PC1 $\approx$ tumor / adjacent-normal and PC2 $\approx$ sex (chr X/Y exclusion rationale); (2) DM diagnostic panel — p-value histogram with permutation-null overlay (visualizing both calibration and signal pervasiveness), paired-$\Delta\beta$ histogram (effect-size biology with biological-significance threshold lines), and $\Delta\beta$-vs-p volcano with five anchor TSGs labelled (VHL, SFRP1, DKK3, GATA5, RASSF1); (3) MEA volcano + responder-fraction histogram; (4) six-locus methylation–expression scatter; (5) VHL/`cg13672843` case study; (6) raw-DM GSEA vs silencing-restricted ORA. Four tables: preprocessing attrition, immune-pathway dropout, FDR sensitivity grid, 18-gene TSG recovery.

Preliminary results (pipeline complete). The central methodological finding: requiring concordant mRNA change in the silencing definition collapses immune-pathway enrichment by **ten orders of magnitude** — HALLMARK_INFLAMMATORY_RESPONSE drops from $p_{adj} = 6 \times 10^{-10}$ in raw-DM GSEA to 1.0 in silencing-restricted ORA, with the entire immune block (Complement, Allograft

Rejection, Interferon γ/α, TNFα NF-κB, IL6 JAK STAT3, KEGG Hematopoietic Cell Lineage) collapsing to non-significance. The remaining silencing-restricted top hits are recognizable cancer biology (EMT, TGF-β, axon guidance, KRAS, KEGG Pathways in Cancer). Supporting numbers: DM yields **222,386 FDR-significant CpGs (69.3%)** with permutation-null $\lambda_{GC}^{\mathrm{perm}} = 0.95$ vs $\lambda_{GC}^{\mathrm{real}} = 36.8$ (visualized directly in the new p-value-histogram diagnostic). The integration filter reduces this to **12,922 silencing (CpG, gene) pairs across 4,483 unique genes**, of which **82.5% are subgroup-driven** ($\hat{\pi}_{\mathrm{resp}} < 0.3$). VHL/`cg13672843` is the textbook subgroup-driven exemplar: cohort-mean $\Delta\beta = 0.04$ (invisible to mean-based DM) but the 32 responder tumors (10.1%) show 33% lower VHL mRNA (Wilcoxon $p = 4.5 \times 10^{-6}$, Cohen's $d = -0.79$). The 18-gene literature TSG panel is recovered with **hypergeometric enrichment of** $2.55\times$ **(**$p = 1.4 \times 10^{-4}$**) for any coupling and** $1.96\times$ **(**$p = 0.014$**) for silencing only** over the 28.3% baseline pass-through rate — statistically significant but modest in magnitude, reflecting the heterogeneous evidence quality of the 18-gene list. External replication on TCGA-KIRP (papillary RCC, 45 paired patients) yields **9,116 silencing pairs across 2,955 genes**, with a 1,889-gene KIRC$\cap$KIRP overlap that is highly enriched over random sampling (Jaccard 0.34, hypergeometric $p < 10^{-300}$).

Expected conclusion. The central methodological contribution is that **silencing — defined as the directional intersection of paired-design differential methylation and within-status methylation–expression association — is sufficient to identify transcriptionally silencing CpGs without an external cell-composition deconvolution panel.** The construct operationalizes Knudson's tumor-suppressor "second hit" in the methylation domain. Its empirical signature: the immune-cell-composition confounding that dominates raw differential-methylation enrichment collapses entirely under the silencing restriction, while recognizable cancer biology (EMT, TGF-β, KRAS) survives — replacing reference-based tools such as EpiDISH or CIBERSORT with a single per-CpG regression. A complementary descriptive observation with broader practical implications: most cancer methylation silencing turns out to be patient-subgroup-specific rather than cohort-wide — VHL, the canonical ccRCC tumor suppressor, exemplifies this — arguing that cancer methylation studies should routinely report per-tumor responder fractions rather than rely on cohort means alone. The entire approach is built from standard statistical primitives; the contribution is *composition* of established building-blocks, not novel inferential machinery.

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